

Conjugation by the flow preserves the spectrum, so the renormalization step is a truncation

The gradient flow is a diffeomorphism of the compact configuration space $G^\mathcal{E}$, so it is implemented by a unitary on $L^2(G^\mathcal{E})$ and conjugation by it leaves every eigenvalue of the Kogut–Susskind Hamiltonian unchanged. Under that conjugation the magnetic term becomes the flowed action, which the flow decreases monotonically, and the electric term becomes the Laplace–Beltrami operator of the pulled-back metric together with a Jacobian potential. In the free case the two effects cancel exactly: the magnetic coefficient of the mode k is multiplied by e^{-2tk^2} and the electric one by e^{+2tk^2} , leaving the frequencies $\omega_k = ck$ untouched. Reading the conjugated Hamiltonian as “classical part plus perturbation” and applying the strong-coupling criterion of the T2 note would give

$$\beta \longmapsto \beta e^{-4tk^2},$$

an arbitrarily large improvement of a criterion whose conclusion, the gap, cannot have changed. The resolution names the content of a renormalization step: **the criterion presupposes a local decomposition, the conjugated electric term is nonlocal with range $\sqrt{8t}$, and any gain is charged entirely to truncating that nonlocality**. A change of variables moves the difficulty and never removes it; the truncation error is the difficulty. This closes the last of the three ways of avoiding the constructive step that this programme has tried, and it states in one line what such a step must control. Constants explicit; nothing promoted.

1. The flow is a unitary change of variables

On a finite lattice the Wilson flow

$$\dot{V}_s(x, \mu) = -g^2 \{ \partial_{x, \mu} S_w(V_s) \} V_s(x, \mu), \quad V_0 = U,$$

has a unique smooth solution for all $s \in \mathbb{R}$ (Lüscher, arXiv:1006.4518v3, equation (1.4) and the sentence following it; passage level via the local companion), so $\Phi_t : U \mapsto V_t$ is a diffeomorphism of the compact manifold $\mathcal{M} = G^\mathcal{E}$ with smooth inverse Φ_{-t} .

Proposition 1. Let $\rho_t = |\det D\Phi_t|$ be the Jacobian of Φ_t with respect to the Haar measure μ . Then

$$(W_t \psi)(U) = \rho_t(U)^{1/2} \psi(\Phi_t(U))$$

is unitary on $L^2(\mathcal{M}, \mu)$, and for any self-adjoint H the operator $W_t H W_t^*$ has the same spectrum, multiplicities and gap.

Proof. Change of variables: for $\psi, \varphi \in L^2$, $\int \overline{(W_t \psi)} (W_t \varphi) d\mu = \int \rho_t \overline{\psi(\Phi_t U)} \varphi(\Phi_t U) d\mu(U) = \int \overline{\psi} \varphi d\mu$ since $\rho_t d\mu = \Phi_t^* d\mu$. Surjectivity follows from $W_t^{-1} = W_{-t}$ up to the Jacobian cocycle. Unitary conjugation preserves the spectrum. $\square$

2. The conjugated Hamiltonian

Write $H = \frac{\hbar c g^2}{2a} K + \frac{2\hbar c}{ag^2} V$ with $K = \sum_\ell (-\Delta_\ell)$ and $V(U) = \sum_p (N - \text{Re tr } U_p)$.

Magnetic term. For a multiplication operator, $W_t M_V W_t^* = M_{V \circ \Phi_t}$, computed directly from the definition: the two Jacobian factors cancel. So the magnetic potential becomes the **flowed action**, and $S_w(V_t)$ is a monotonically decreasing function of t (Lüscher, immediately after (1.4)), whence

$$0 \leq V \circ \Phi_t \leq V.$$

Electric term. K is the Laplace–Beltrami operator of the product bi-invariant metric on $\mathcal{M}$. Conjugation by W_t gives the Laplace–Beltrami operator of the pulled-back metric $\Phi_t^* g$, together with a first-order term and a potential built from ρ_t and its derivatives:

$$W_t K W_t^* = -\Delta_{\Phi_t^* g} + Q_t, \quad Q_t = \rho_t^{1/2} (\Delta_{\Phi_t^* g} \rho_t^{-1/2}),$$

the standard ground-state transform. The pulled-back metric is no longer a product over links: $D\Phi_t$ mixes links within the flow radius $\sqrt{8t}$, so the conjugated kinetic operator has **range** $\sqrt{8t}$.

3. The free case: exact cancellation

For the abelian theory in Coulomb gauge the flow is the heat equation (as in the free-field note §2), so Φ_t is the linear map $A \mapsto e^{t\Delta}A$, its Jacobian is a constant, and W_t is the induced unitary with no potential term. The conjugated Hamiltonian is, mode by mode,

$$W_t H W_t^* = \int \frac{d^3k}{(2\pi)^3} \left[\frac{g^2 c}{2\hbar} e^{+2tk^2} |\hat{\Pi}(k)|^2 + \frac{\hbar c}{2g^2} k^2 e^{-2tk^2} |\hat{A}(k)|^2 \right],$$

since the conjugate momentum transforms by the inverse transpose of the map on the fields. Each mode is an oscillator of frequency

$$\omega_k = \sqrt{(c e^{+2tk^2})(c k^2 e^{-2tk^2})} = ck,$$

independent of t , as Proposition 1 requires. The flow makes the potential smaller and the kinetic term correspondingly stiffer, and the product is invariant.

4. What this does to a locality-based criterion

The criterion of the T2 note compares the norm of the perturbation with the local gap of the classical part: $\beta = \|\phi\|/(\text{local gap})$. Applied to the conjugated Hamiltonian, mode by mode in the free case, the magnetic norm carries e^{-2tk^2} and the local gap of the kinetic term carries e^{+2tk^2} , so

$$\beta \longmapsto \beta e^{-4tk^2},$$

which can be made as small as desired by taking t large. On a lattice the modes run to $k \sim \pi/a$, so at flow time $t = (Ma)^2$ the improvement factor reaches $e^{-4\pi^2 M^2}$. Since the spectrum is unchanged, no statement about the gap can follow from this improvement.

Where the criterion loses its validity. Yarotsky's hypotheses require the classical part to be a sum of single-site terms and the perturbation to have finite range. After conjugation, the kinetic operator has range $\sqrt{8t}$ and the decomposition into single-link terms is gone. The apparent gain is therefore the price of dropping a hypothesis, and a criterion that keeps the hypothesis must first **truncate** the conjugated kinetic operator back to finite range. Truncation changes the operator, and the resulting gap bound applies to the truncated Hamiltonian; its relation to the original is exactly the truncation error.

Corollary 2. Any gap criterion formulated through a local decomposition into classical part and small perturbation fails to be invariant under unitary changes of variables, and can be improved without limit by conjugation. Consequently the improvement obtained in a renormalization step is charged in full to the truncation that restores locality, and no change of variables by itself improves a spectral bound.

5. The three closed routes

This completes a set. Of the ways to reach $m > 0$ while avoiding the constructive step, the notes have now closed:

route	why it closes	note
expansion around the free theory	the bound reaches m only at the confinement scale, where the coefficient is nonperturbative	free field §7
real-space blocking by projection	boundary grows like M^2 , block gap does not grow; criterion worsens by $3M^2/8$	blocking criterion
change of variables by the flow	spectrum is invariant, so the gain is the truncation error	this note
upper bounds of any kind	finite negative moments are compatible with $m = 0$	moment hierarchy §5

Each was closed by a computation with explicit constants rather than by an appeal to difficulty, and together they say the same thing: the only place a proof of $m > 0$ can come from is a controlled truncation, applied a finite number of times, with the error at each step small compared with the gap at that scale.

6. What a truncation must satisfy

The requirement can now be stated with the constants of this programme. Let $H^{(n)}$ be the theory at scale $\ell_n = 2^n a$, with gap $\Delta_n \geq \kappa \hbar c g_n^2 / \ell_n$ in the regime where the criterion applies, and let T_n be the truncation to range ℓ_n following the conjugation. A proof needs

$$\|H^{(n)} - T_n W_{t_n} H^{(n)} W_{t_n}^* T_n^*\|_{\text{relative}} \ll \Delta_n \quad \text{for each of the } n \simeq \frac{1}{2b_0 \log 2} \left(\frac{1}{g_{\text{UV}}^2} - \frac{1}{g_{\text{thr}}^2} \right)$$

steps, with the norm taken in the relative sense of the Schur note Lemma 1', so that constant shifts are free. The flow radius $\sqrt{8t_n}$ must be of order ℓ_n for the conjugation to reduce the action at the right scale, and the truncation error is governed by the decay of the flow Jacobian $D\Phi_{t_n}$ beyond that radius, which for the free case is Gaussian in $|x|/\sqrt{8t_n}$ and in the interacting case is the object the construction must bound.

7. Consequence for STATE

The three ways of avoiding the constructive step are closed, and the step itself is stated in the form the constants of this programme give it: a conjugation by the flow at radius ℓ_n followed by a truncation whose relative error must stay below the gap at that scale, iterated a finite number of times. The next tractable item is the decay of the flow Jacobian $D\Phi_t$ on the lattice, which controls the truncation error and which the free case fixes exactly; that is a bounded question about the flow equation and is the natural next note.